

Picup System & AI — Enterprise Production Roadmap

This document outlines the complete enterprise roadmap for the Picup System & AI platform, including frontend architecture, backend APIs, AI integration, driver automation, production deployment, monitoring, security, and rollout phases.

1. Frontend Development Phase

- Build the complete operational web interface.
 - Use React + TailwindCSS for scalable UI architecture.
 - Implement Firebase Authentication and role-based access.
 - Create KPI dashboards, training systems, chat modules, and AI assistant interfaces.
 - Enable realtime updates using Firebase SDK and Realtime Database.
-

2. Authentication Module

- Google login integration.
 - Restrict access to Picup email domains.
 - Implement RBAC (Role-Based Access Control).
 - Admin session handling and route protection.
 - Support roles including Admin, HR, QA, Team Leaders, Finance, Employees, and Trainers.
-

3. Dashboard Module

- KPI dashboards.
 - Attendance tracking.
 - Calls made and notes logged.
 - SLA performance metrics.
 - Live operational analytics.
 - Daily, weekly, monthly, and 4-month trend charts.
-

4. Training & SOP Module

- Upload SOPs and training documents.

- Generate AI-based assessments.
 - Timed exams with auto marking.
 - Results dashboard and refresher triggers.
 - Automated email alerts for failed assessments.
-

5. Picup Slack Communication Module

- Group chats and direct messaging.
 - Voice and video calling.
 - File uploads and emoji support.
 - Scheduled messaging and AI auto replies.
 - Message editing and deletion.
-

6. Phila AI Live Chat

- AI-powered driver assistance.
 - Automatic trip removal.
 - SLA breach detection.
 - Finance support queries.
 - AI SOP knowledge search.
 - Driver suspension automation.
-

7. Backend API Architecture

- Build APIs using Node.js and Express.js.
 - Use Firebase Functions and Cloud Run for scalability.
 - Implement authentication, KPI, training, finance, driver, and chat APIs.
 - Enable realtime cloud processing using Cloud Tasks.
-

8. Firebase Infrastructure

- Firebase Authentication.
- Firestore Database.
- Firebase Hosting.
- Firebase Storage.
- Cloud Functions.
- Crashlytics and Analytics integration.

9. AI Integration Architecture

- Gemini Pro for enterprise AI.
 - Gemini Flash for fast operational responses.
 - Embeddings and vector search for SOP retrieval.
 - OCR extraction for uploaded documents.
 - Retrieval-Augmented Generation (RAG) workflows.
-

10. Driver Automation Engine

- GPS monitoring.
 - Distance validation.
 - Late collection warnings.
 - Auto suspension workflows.
 - Trip lifecycle automation.
 - Route deviation detection.
-

11. DevOps & CI/CD

- GitHub Actions deployment pipelines.
 - Automated testing and security scanning.
 - Firebase deployment automation.
 - Production monitoring and infrastructure logging.
-

12. Security & Compliance

- Firestore security rules.
 - JWT validation.
 - HTTPS-only architecture.
 - Role-based authorization.
 - MFA for administrators.
 - Audit logging and activity tracking.
-

13. Testing Strategy

- Unit testing with Jest.

- API testing with Supertest.
 - E2E testing with Cypress.
 - Load testing using k6.
 - Security testing using OWASP ZAP.
-

14. Enterprise Microservices

- Authentication Service.
 - KPI Service.
 - Training Service.
 - AI Service.
 - Driver Automation Service.
 - Finance Service.
 - Notification Service.
 - Audit Logging Service.
-

15. Production Rollout Phases

- Phase 1 — Infrastructure Setup.
 - Phase 2 — Frontend Development.
 - Phase 3 — Training System.
 - Phase 4 — AI Integration.
 - Phase 5 — Driver Automation.
 - Phase 6 — Communication Platform.
 - Phase 7 — Finance Automation.
 - Phase 8 — Production Launch.
-

16. Recommended Team Structure

- Frontend Developers.
 - Backend Engineers.
 - Firebase Engineers.
 - AI Engineers.
 - QA Team.
 - DevOps Team.
 - Product Owner.
-

17. Final Production Architecture

- Frontend Application.
 - Firebase Hosting.
 - Authentication Layer.
 - Firestore Database.
 - Cloud Functions.
 - Gemini AI Services.
 - Driver Application.
 - Operational Dashboards.
-

Estimated Enterprise Timeline

Phase	Estimated Duration
Infrastructure Setup	2 Weeks
Frontend Development	3 Weeks
Training System	3 Weeks
AI Integration	2 Weeks
Driver Automation	4 Weeks
Communication Platform	4 Weeks
Finance Automation	2 Weeks
Production Launch	2 Weeks

The Picup System & AI platform is designed as a fully integrated enterprise operational ecosystem. The architecture combines AI-driven automation, realtime logistics intelligence, employee performance management, communication systems, finance workflows, and scalable cloud infrastructure into a single production-grade platform.